

## ACTIVITY 2

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### Scenario 1: Smart City Air Quality Monitoring

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CODE: python

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

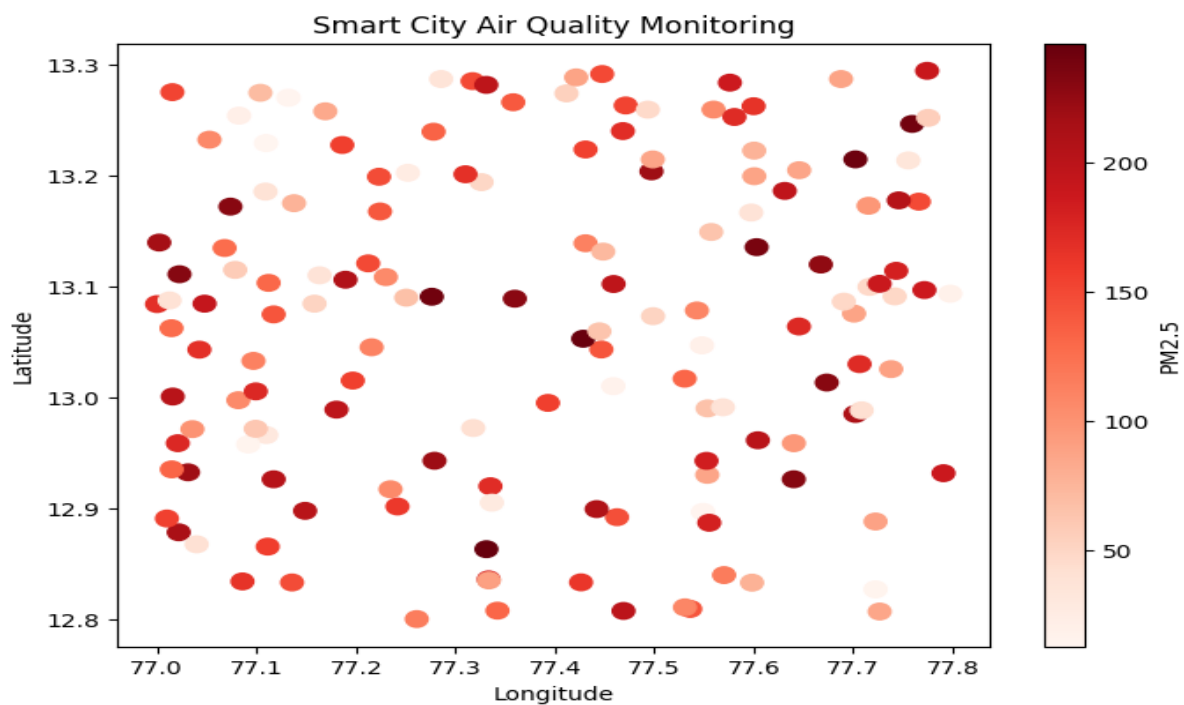
np.random.seed(1)

n = 150
df = pd.DataFrame({
    "Longitude": np.random.uniform(77.0, 77.8, n),
    "Latitude": np.random.uniform(12.8, 13.3, n),
    "PM2.5": np.random.randint(10, 250, n)
})

plt.figure(figsize=(8,6))
plt.scatter(df["Longitude"], df["Latitude"],
            c=df["PM2.5"], cmap="Reds", s=80)

plt.colorbar(label="PM2.5")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.title("Smart City Air Quality Monitoring")
plt.show()
```

OUTPUT:



## Scenario 1: Smart City Air Quality Monitoring

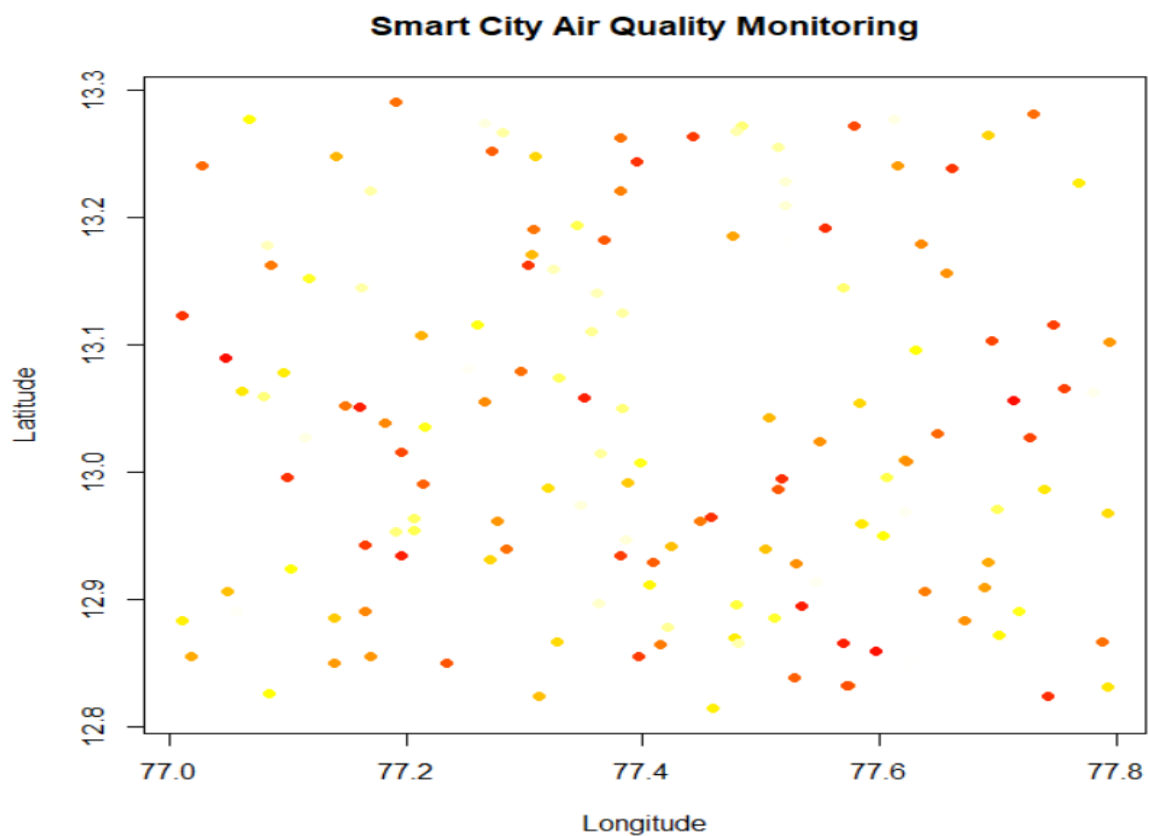
### CODE: R program

```
set.seed(1)

n <- 150
longitude <- runif(n, 77.0, 77.8)
latitude <- runif(n, 12.8, 13.3)
pm25 <- sample(10:250, n, replace=TRUE)

plot(longitude, latitude,
      col=heat.colors(250)[pm25],
      pch=19,
      xlab="Longitude",
      ylab="Latitude",
      main="Smart City Air Quality Monitoring")
```

### OUTPUT:



## 2. Scenario 2: Earthquake Risk Analysis

### CODE: Python

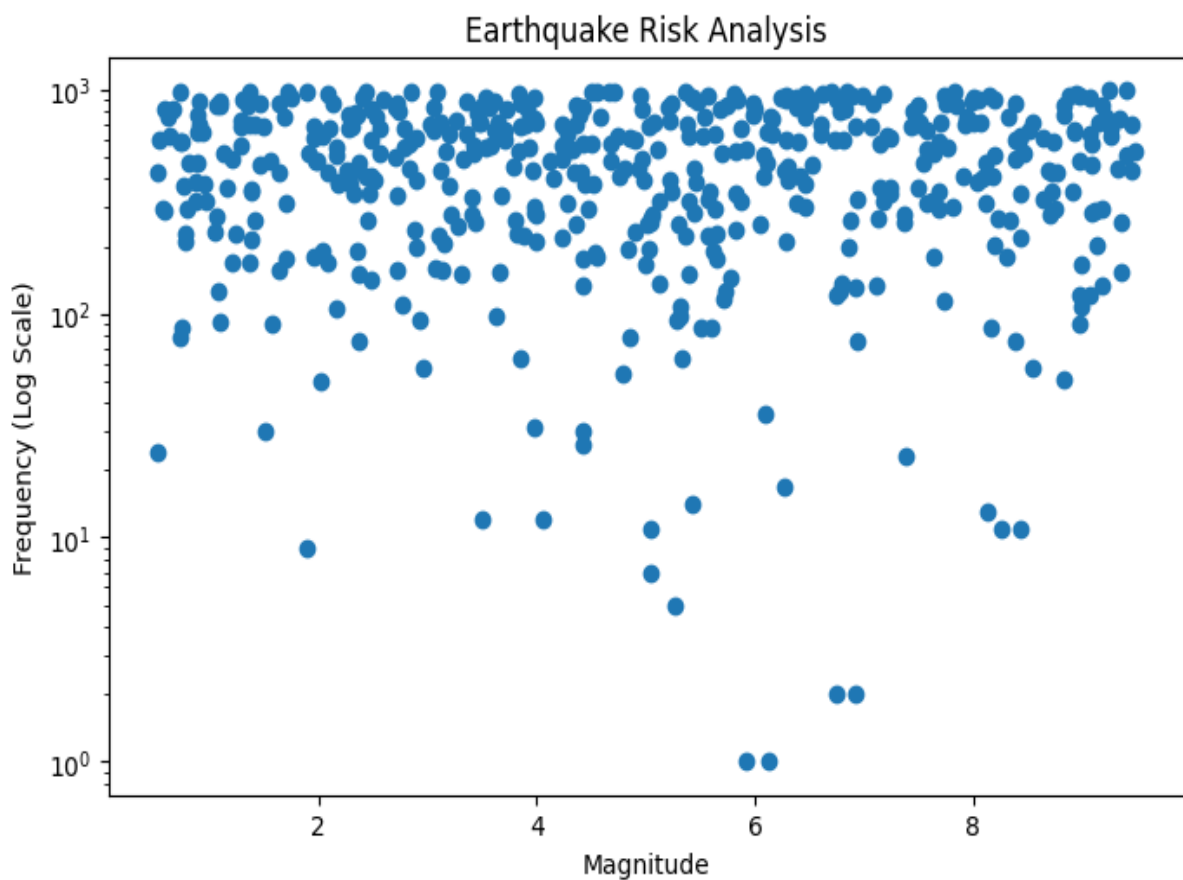
```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(2)

magnitude = np.random.uniform(0.5,9.5,500)
frequency = np.random.randint(1,1000,500)

plt.figure(figsize=(8,5))
plt.scatter(magnitude,frequency)
plt.yscale("log")
plt.xlabel("Magnitude")
plt.ylabel("Frequency (Log Scale)")
plt.title("Earthquake Risk Analysis")
plt.show()
```

### OUTPUT:



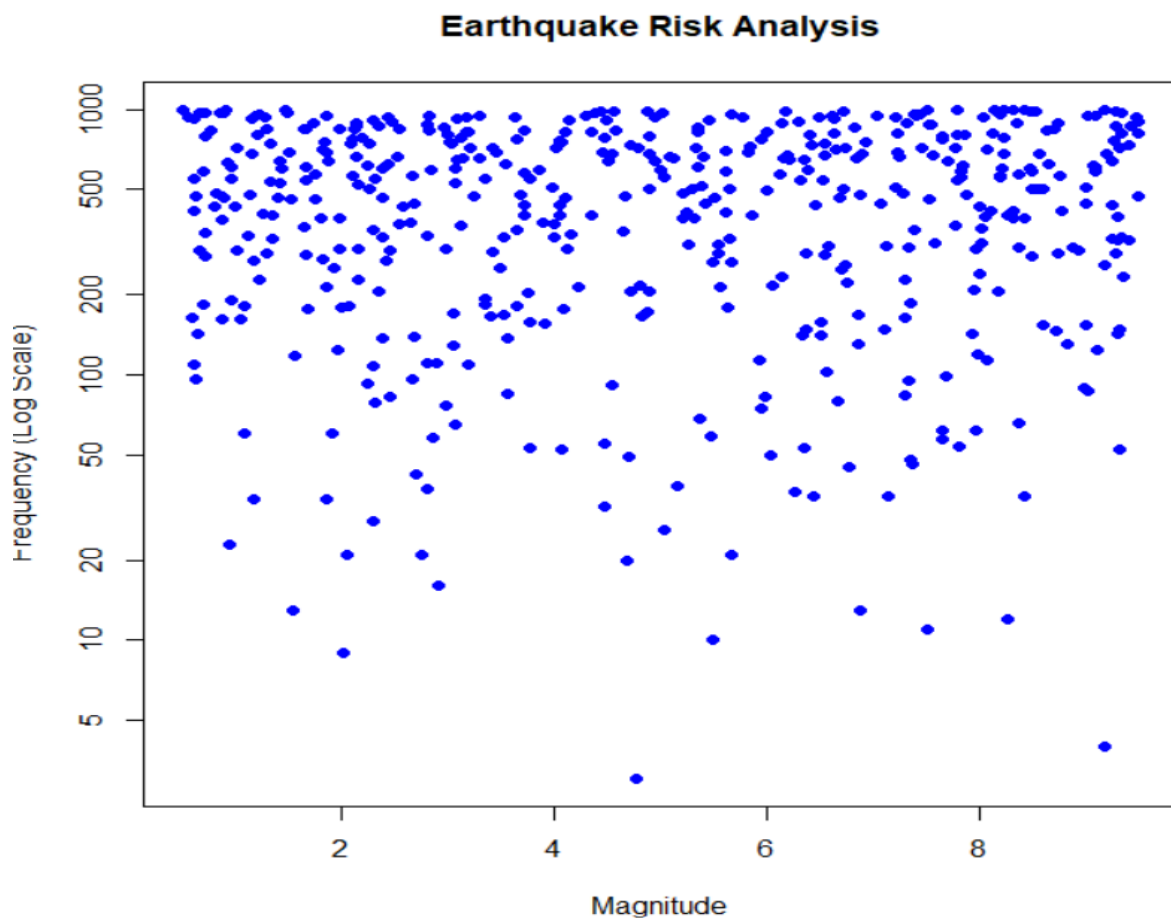
## CODE: R program

```
set.seed(2)

magnitude <- runif(500,0.5,9.5)
frequency <- sample(1:1000,500,replace=TRUE)

plot(magnitude, frequency,
     log="y",
     pch=19,
     col="blue",
     xlab="Magnitude",
     ylab="Frequency (Log Scale)",
     main="Earthquake Risk Analysis")
```

## OUTPUT:



### Scenario 3: Wind Direction Analysis (Polar Plot)

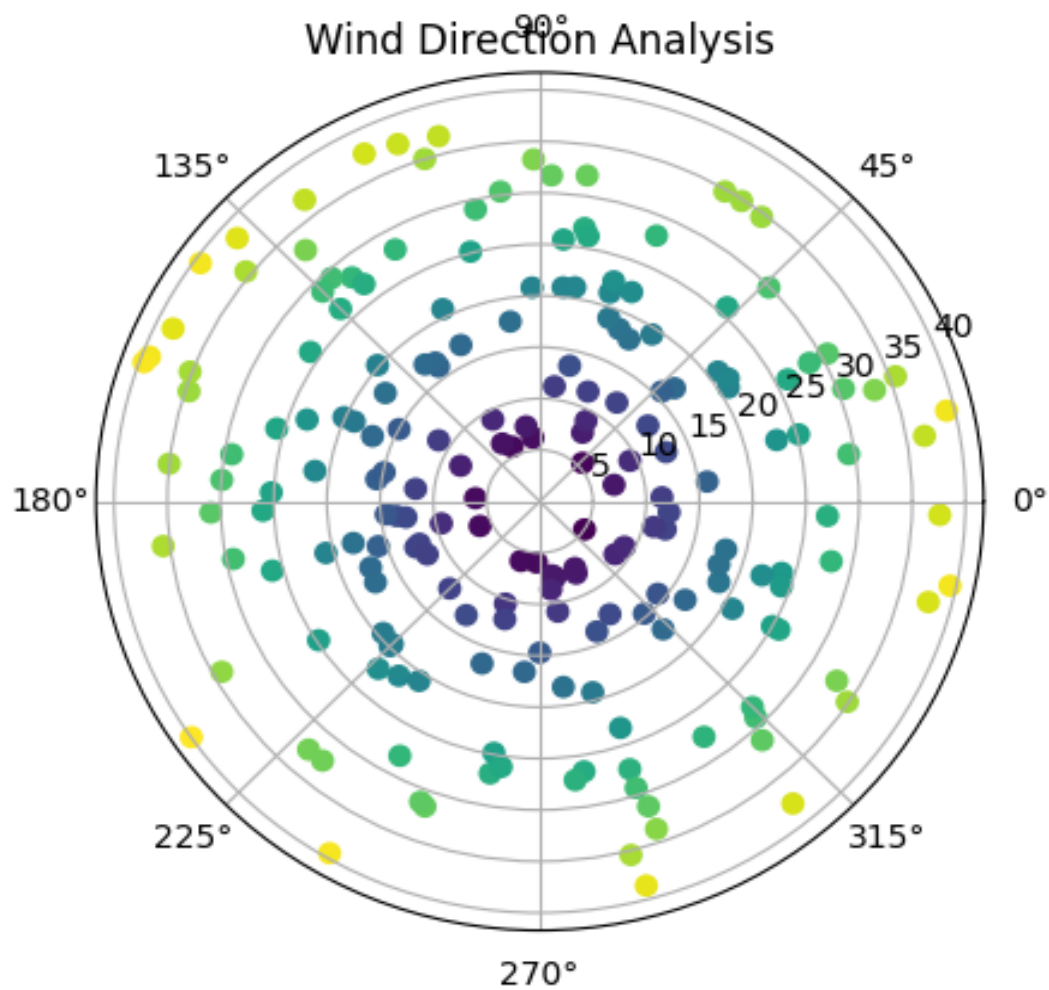
#### CODE: Python

```
import numpy as np
import matplotlib.pyplot as plt

angles=np.deg2rad(np.random.randint(0,360,200))
speed=np.random.uniform(5,40,200)

ax=plt.subplot(111,projection='polar')
ax.scatter(angles,speed,c=speed,cmap="viridis")
plt.title("Wind Direction Analysis")
plt.show()
```

#### OUTPUT:



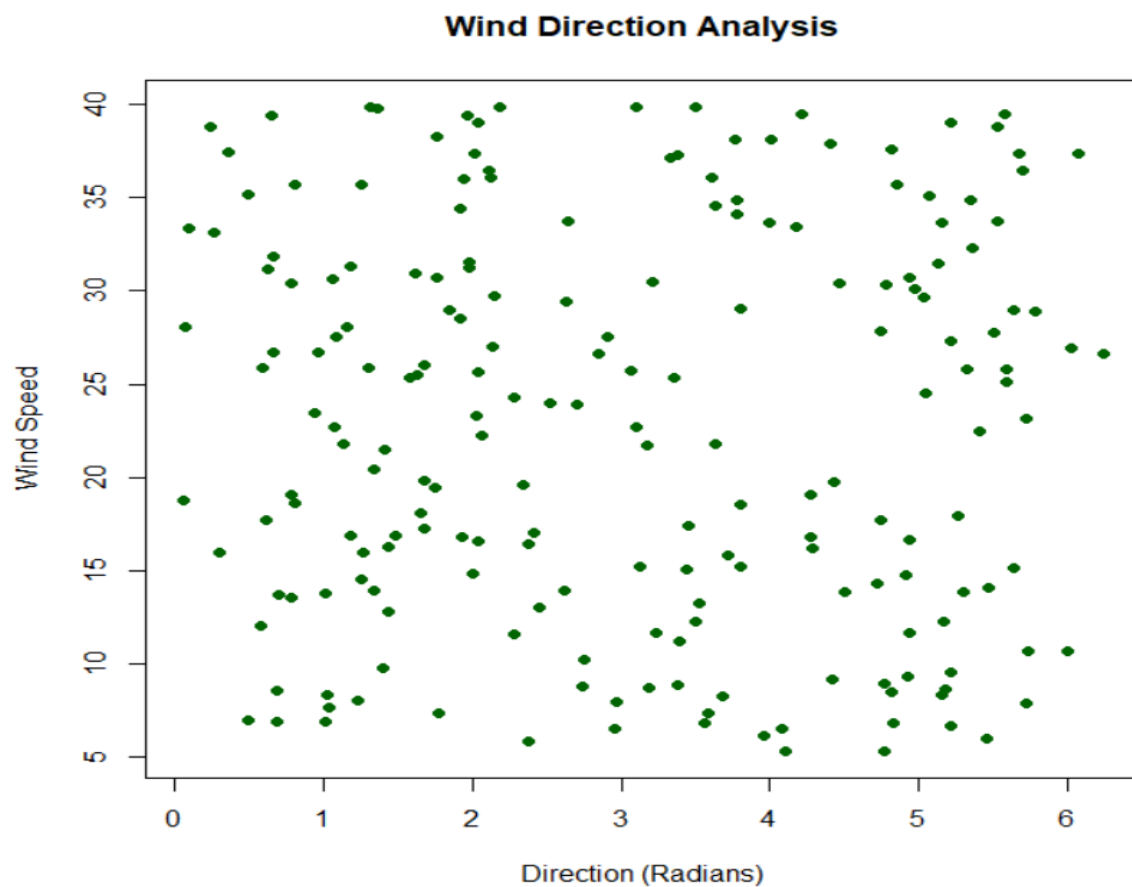
## CODE: R program

```
set.seed(3)

direction <- runif(200,0,2*pi)
speed <- runif(200,5,40)

plot(direction, speed,
      main="Wind Direction Analysis",
      xlab="Direction (Radians)",
      ylab="Wind Speed",
      col="darkgreen",
      pch=19)
```

## OUTPUT:



## Scenario 4: Hospital Patient Monitoring Dashboard

### CODE: Python

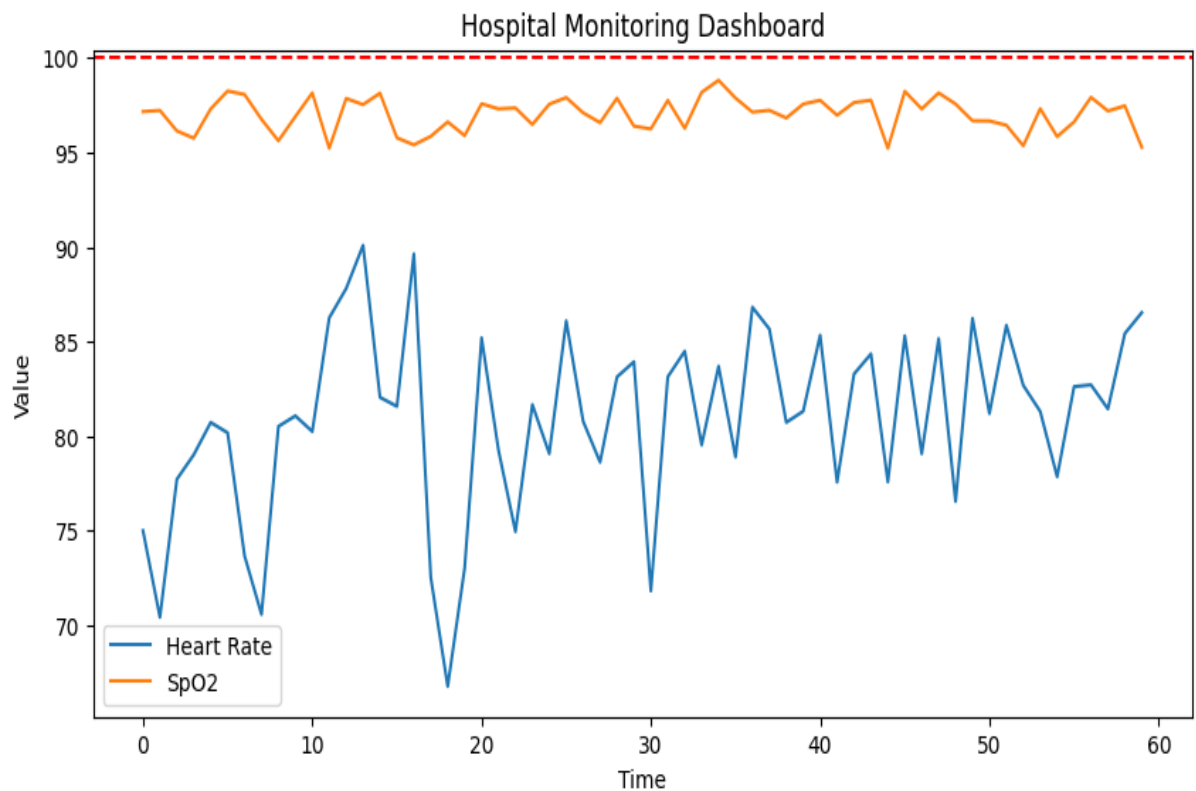
```
import numpy as np
import matplotlib.pyplot as plt

time=np.arange(60)

heart=np.random.normal(80,5,60)
spo2=np.random.normal(97,1,60)

plt.figure(figsize=(10,5))
plt.plot(time,heart,label="Heart Rate")
plt.plot(time,spo2,label="SpO2")
plt.axhline(100,color='r',linestyle='--')
plt.legend()
plt.xlabel("Time")
plt.ylabel("Value")
plt.title("Hospital Monitoring Dashboard")
plt.show()
```

### OUTPUT:



## CODE: R program

```
set.seed(4)

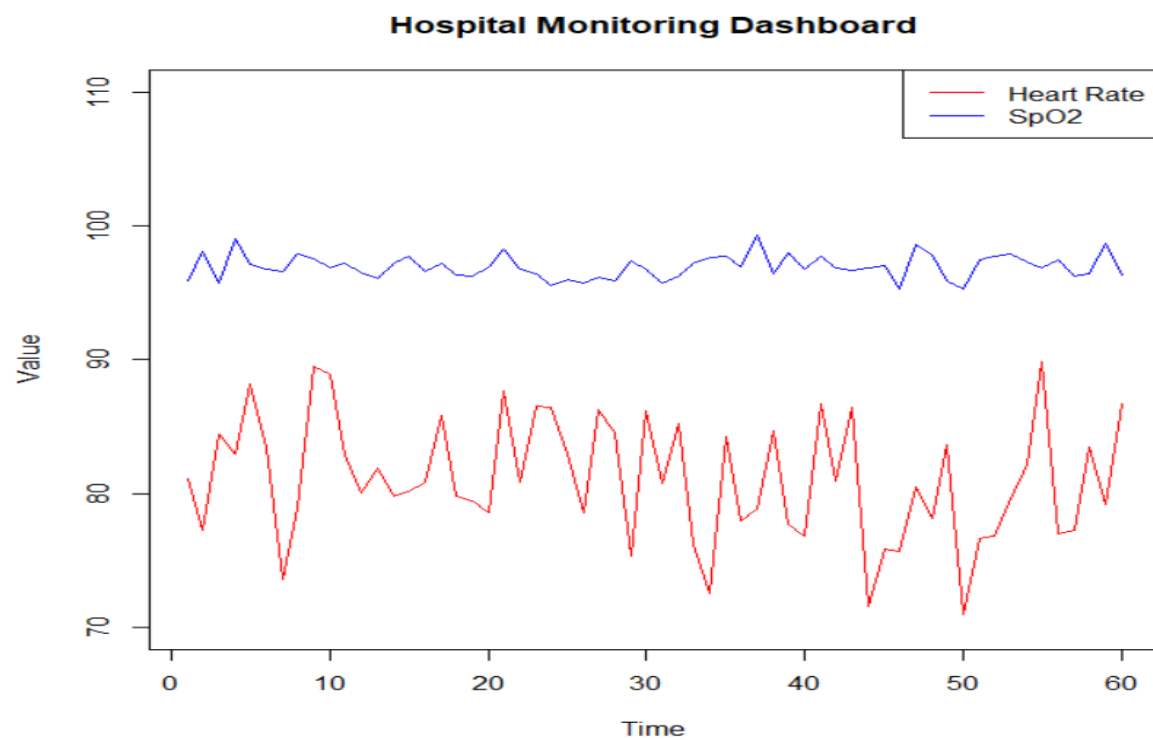
time <- 1:60
heart <- rnorm(60,80,5)
spo2 <- rnorm(60,97,1)

plot(time, heart,
      type="l",
      col="red",
      ylim=c(70,110),
      xlab="Time",
      ylab="Value",
      main="Hospital Monitoring Dashboard")

lines(time, spo2, col="blue")

legend("topright",
      legend=c("Heart Rate","SpO2"),
      col=c("red","blue"),
      lty=1)
```

## OUTPUT:



## Scenario 5: Global Climate Change Study (Heatmap)

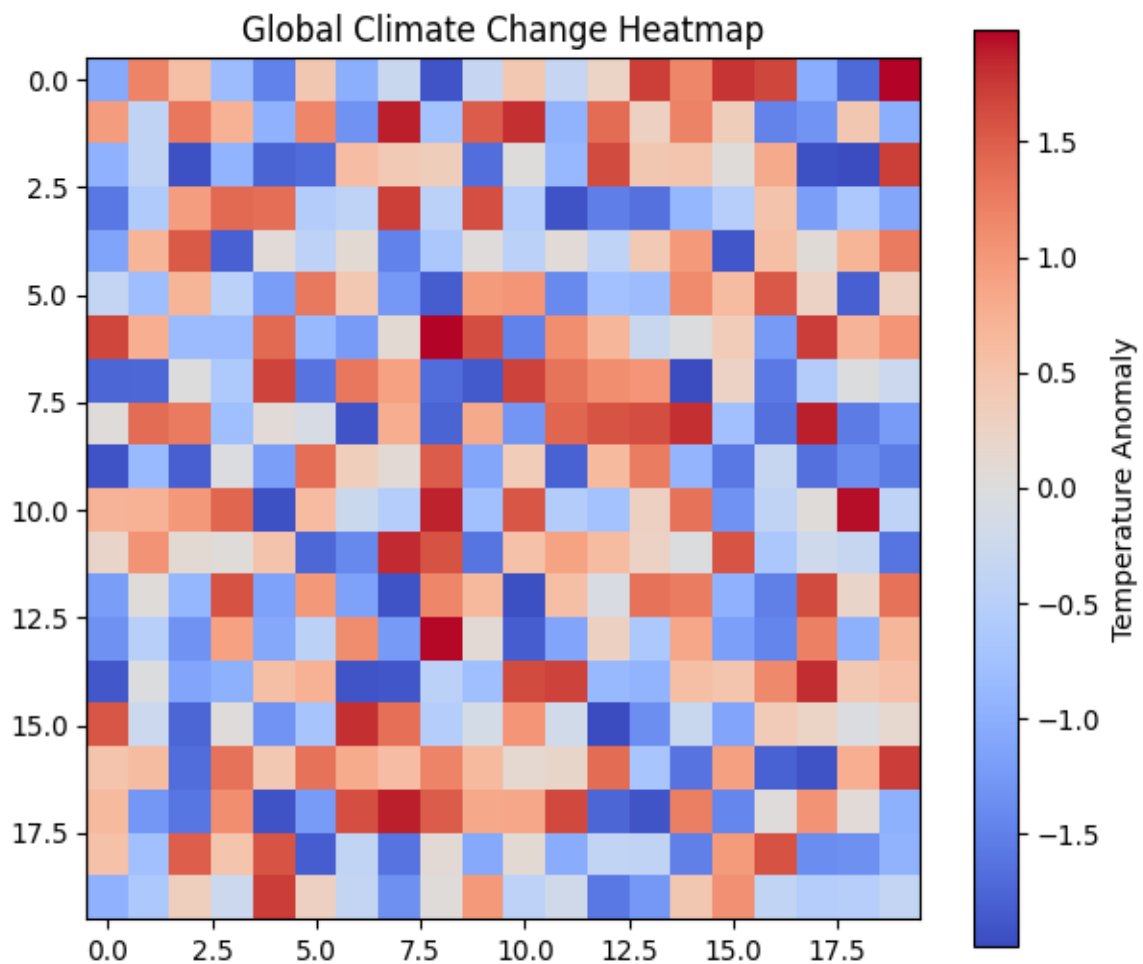
### CODE: Python

```
import numpy as np
import matplotlib.pyplot as plt

data=np.random.uniform(-2,2,(20,20))

plt.figure(figsize=(7,6))
plt.imshow(data,cmap="coolwarm")
plt.colorbar(label="Temperature Anomaly")
plt.title("Global Climate Change Heatmap")
plt.show()
```

### OUTPUT:



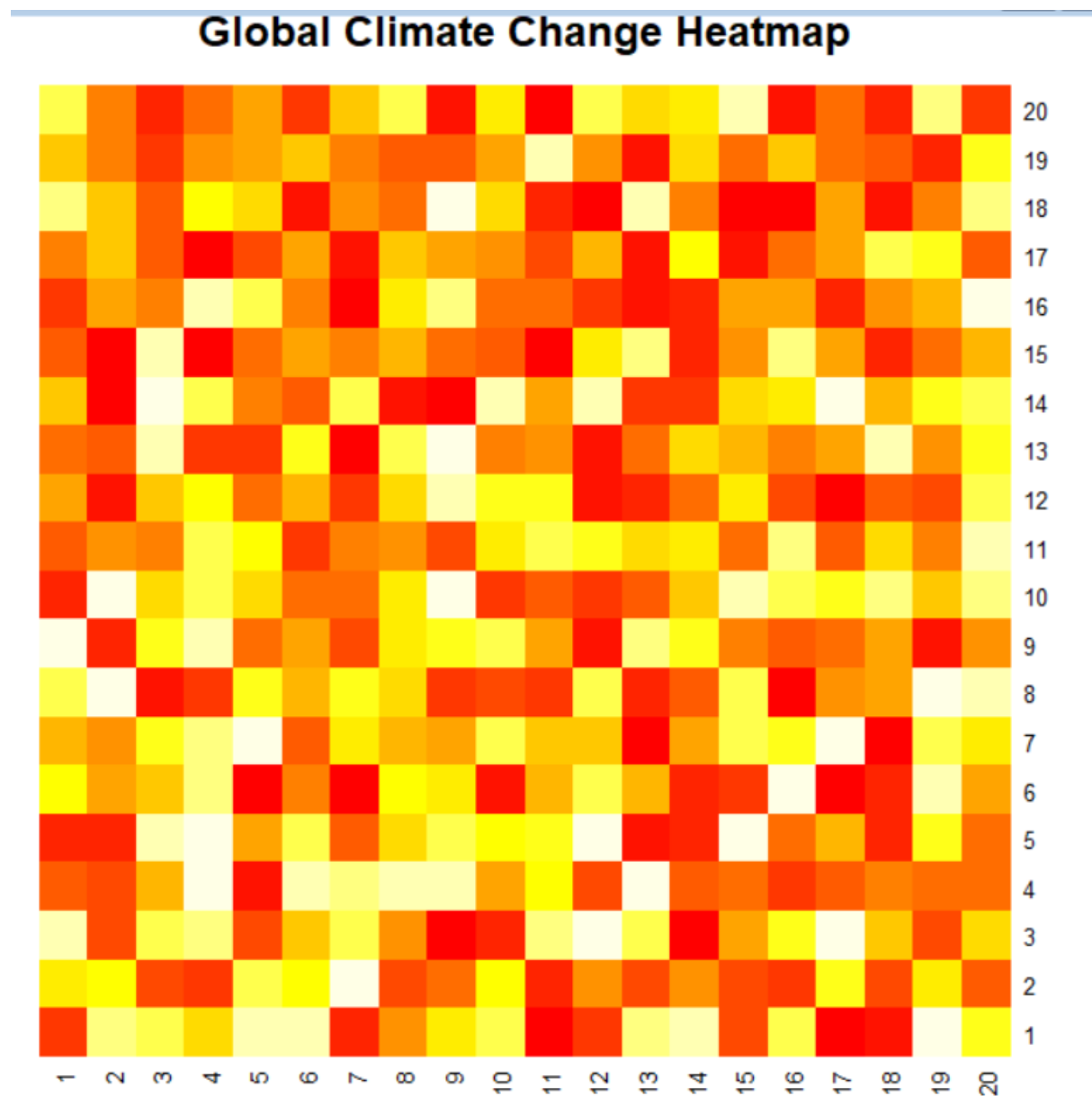
## CODE: R program

```
set.seed(5)

data <- matrix(runif(400,-2,2),20,20)

heatmap(data,
        Colv=NA,
        Rowv=NA,
        scale="none",
        col=heat.colors(20),
        main="Global Climate Change Heatmap")
```

## OUTPUT:



## Scenario 6: E-Commerce Sales Analytics

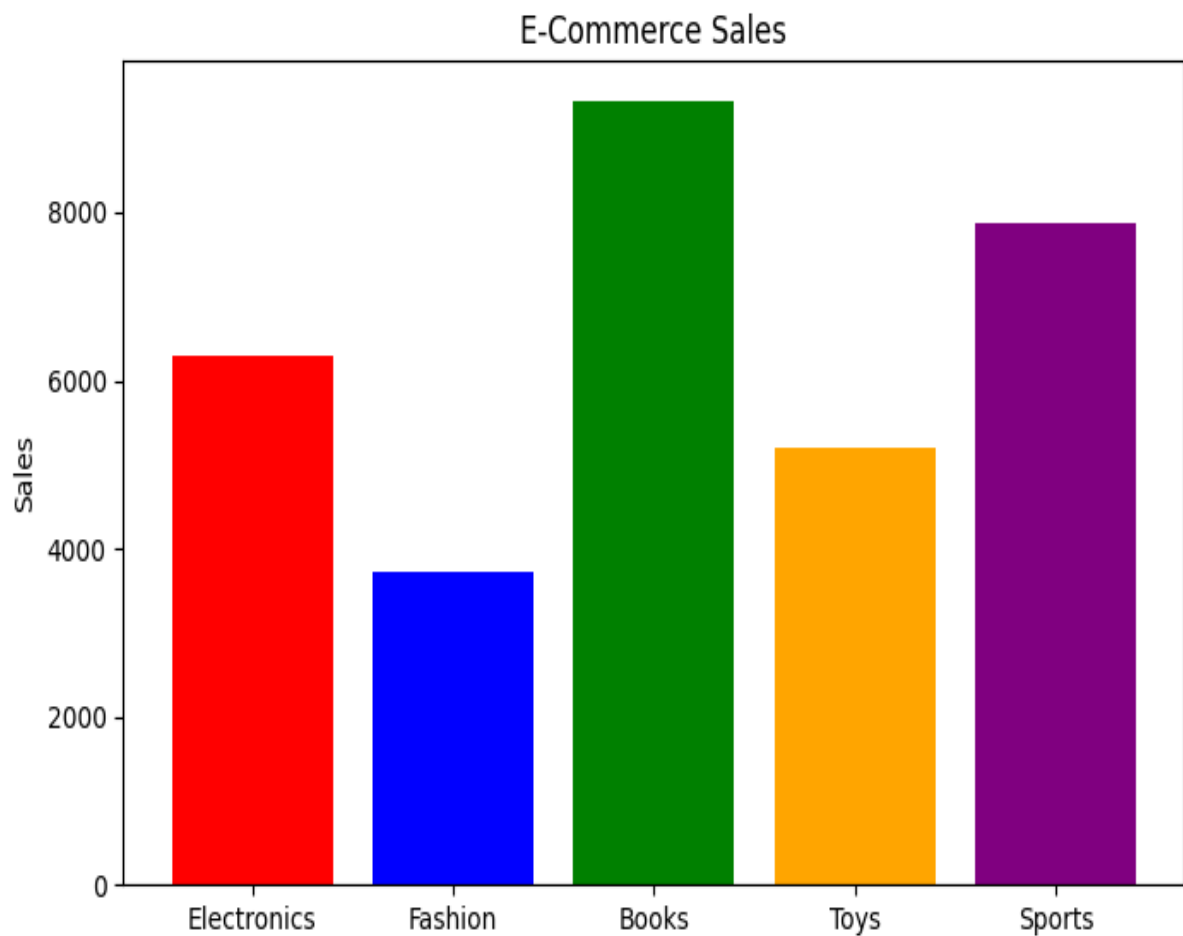
### CODE: Python

```
import numpy as np
import matplotlib.pyplot as plt

categories=["Electronics","Fashion","Books","Toys","Sports"]
sales=np.random.randint(1000,10000,5)

plt.figure(figsize=(8,5))
plt.bar(categories,sales,color=["red","blue","green","orange","purple"]
)
plt.title("E-Commerce Sales")
plt.ylabel("Sales")
plt.show()
```

### OUTPUT:

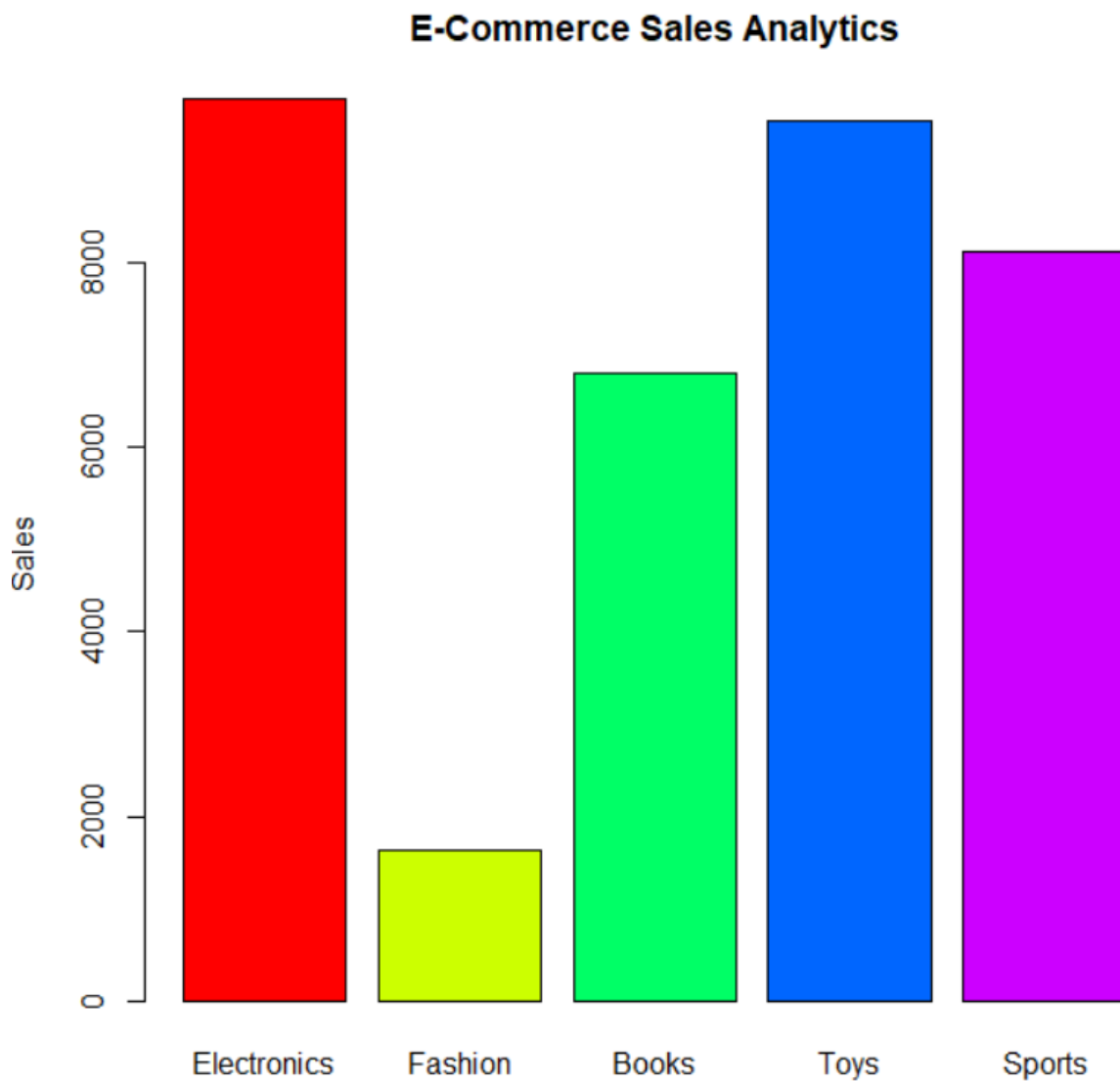


## CODE: R program

```
categories <- c("Electronics","Fashion","Books","Toys","Sports")
sales <- sample(1000:10000,5)

barplot(sales,
        names.arg=categories,
        col=rainbow(5),
        main="E-Commerce Sales Analytics",
        ylab="Sales")
```

## OUTPUT:



## Scenario 7: Satellite Image Vegetation Analysis

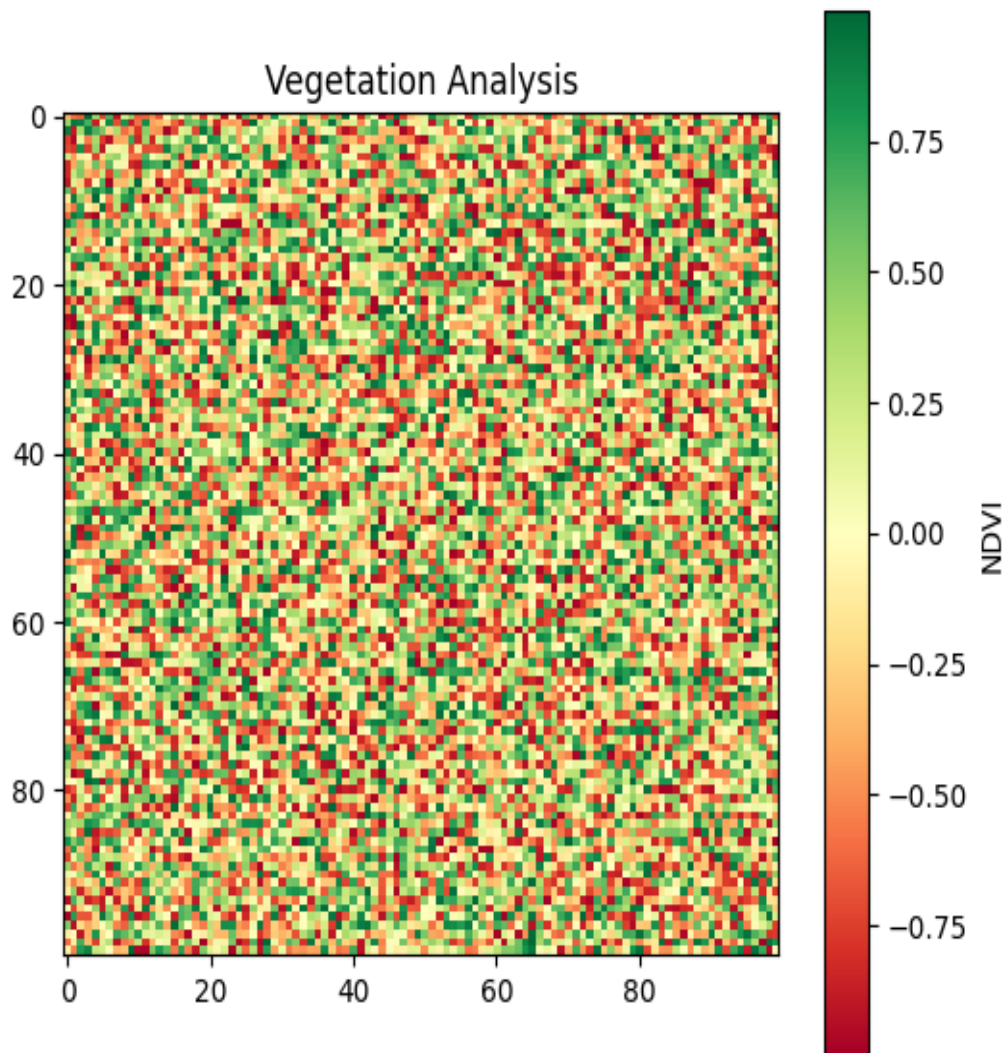
### CODE: Python

```
import numpy as np
import matplotlib.pyplot as plt

ndvi=np.random.uniform(-1,1,(100,100))

plt.figure(figsize=(6,6))
plt.imshow(ndvi,cmap="RdYlGn")
plt.colorbar(label="NDVI")
plt.title("Vegetation Analysis")
plt.show()
```

### OUTPUT:



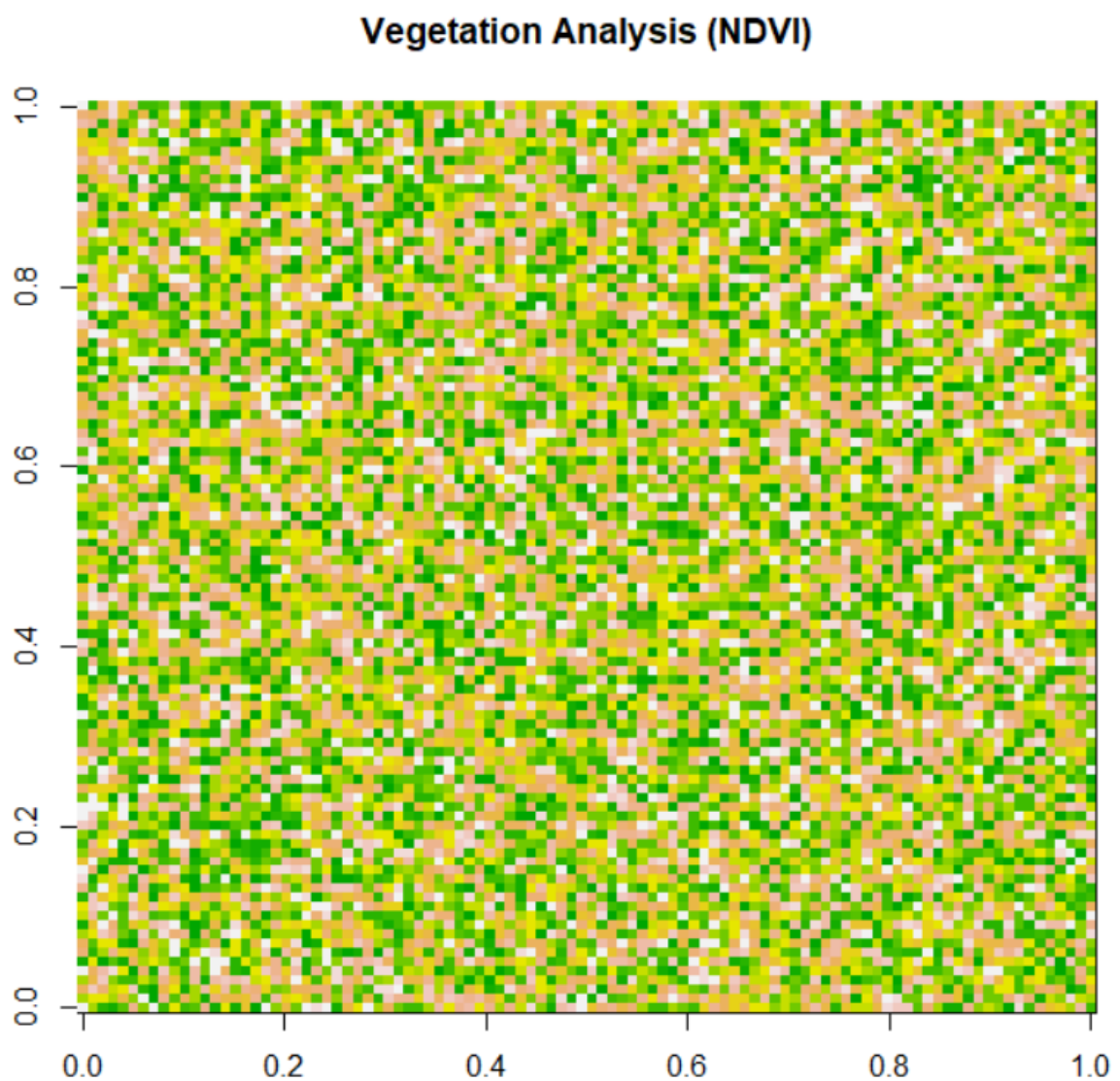
## CODE: R program

```
set.seed(7)

ndvi <- matrix(runif(10000,-1,1),100,100)

image(ndvi,
      col=terrain.colors(20),
      main="Vegetation Analysis (NDVI)")
```

## OUTPUT:



## Scenario 8: Cryptocurrency Market Volatility

### CODE: Python

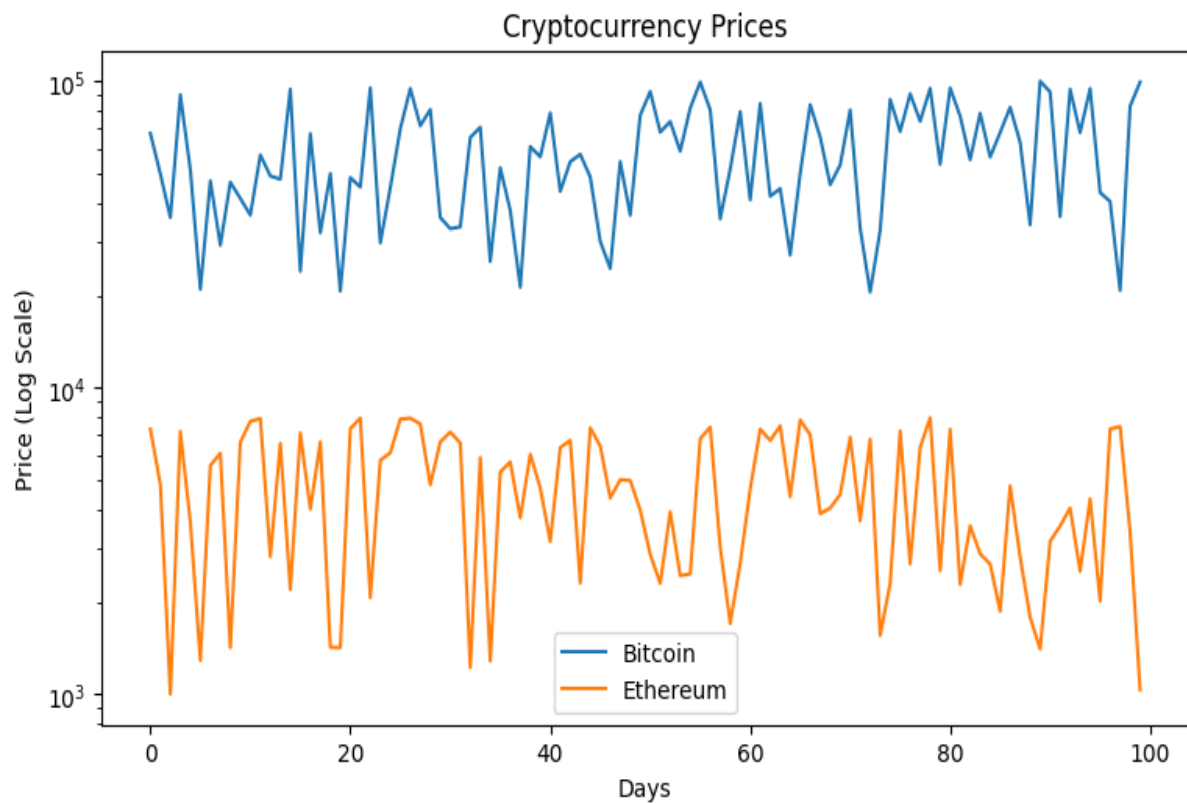
```
import numpy as np
import matplotlib.pyplot as plt

days=np.arange(100)

btc=np.random.uniform(20000,100000,100)
eth=np.random.uniform(1000,8000,100)

plt.figure(figsize=(9,5))
plt.plot(days,btc,label="Bitcoin")
plt.plot(days,eth,label="Ethereum")
plt.yscale("log")
plt.legend()
plt.title("Cryptocurrency Prices")
plt.xlabel("Days")
plt.ylabel("Price (Log Scale)")
plt.show()
```

### OUTPUT:



## CODE: Rprogram

```
set.seed(8)

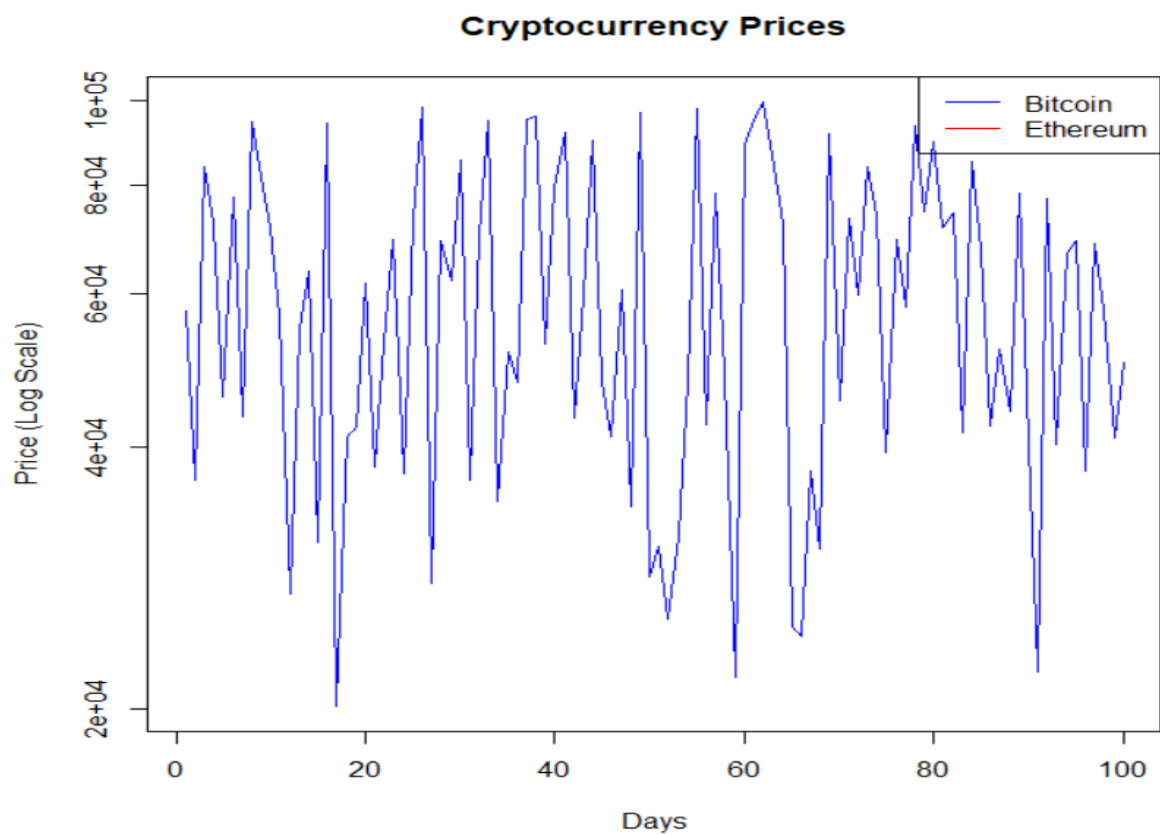
days <- 1:100
btc <- runif(100,20000,100000)
eth <- runif(100,1000,8000)

plot(days, btc,
      type="l",
      log="y",
      col="blue",
      xlab="Days",
      ylab="Price (Log Scale)",
      main="Cryptocurrency Prices")

lines(days, eth, col="red")

legend("topright",
      legend=c("Bitcoin","Ethereum"),
      col=c("blue","red"),
      lty=1)
```

## OUTPUT:



## Scenario 9: Marathon Runner Performance Analysis

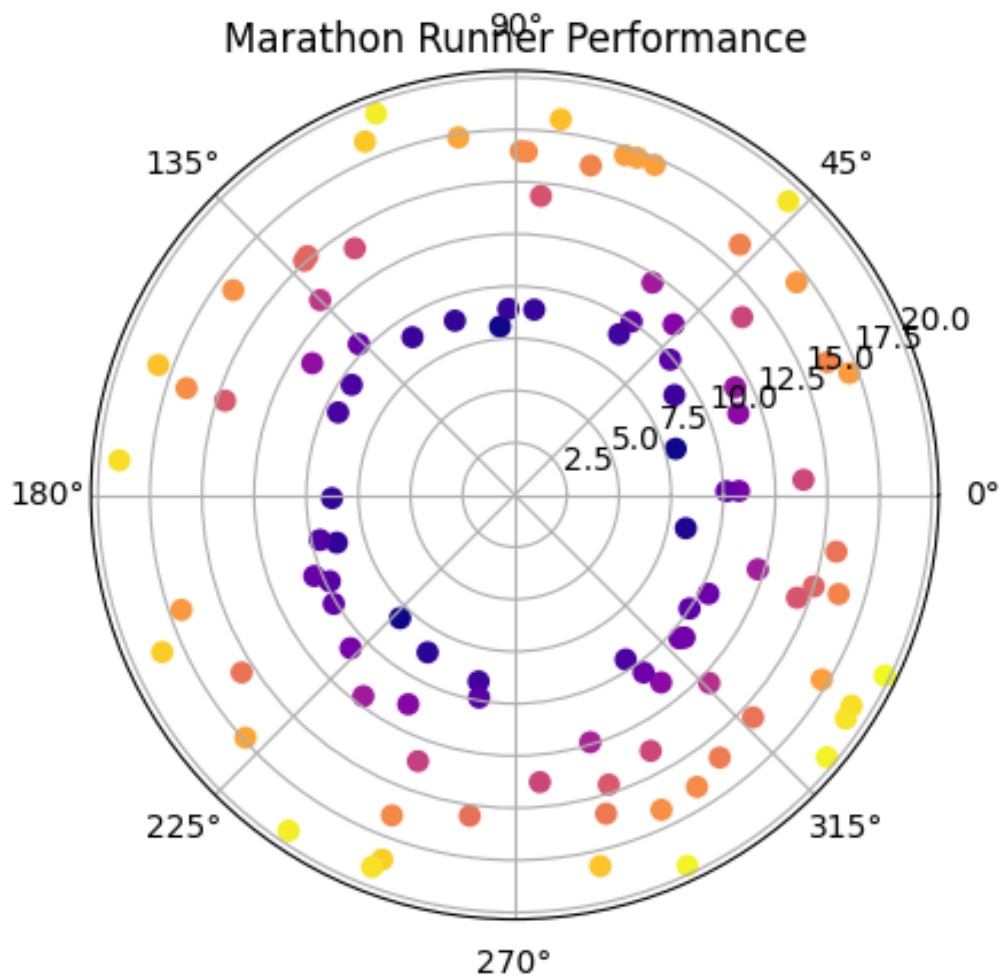
### CODE: Python

```
import numpy as np
import matplotlib.pyplot as plt

direction=np.deg2rad(np.random.randint(0,360,100))
speed=np.random.uniform(8,20,100)

ax=plt.subplot(111,projection="polar")
ax.scatter(direction,speed,c=speed,cmap="plasma")
plt.title("Marathon Runner Performance")
plt.show()
```

### OUTPUT:



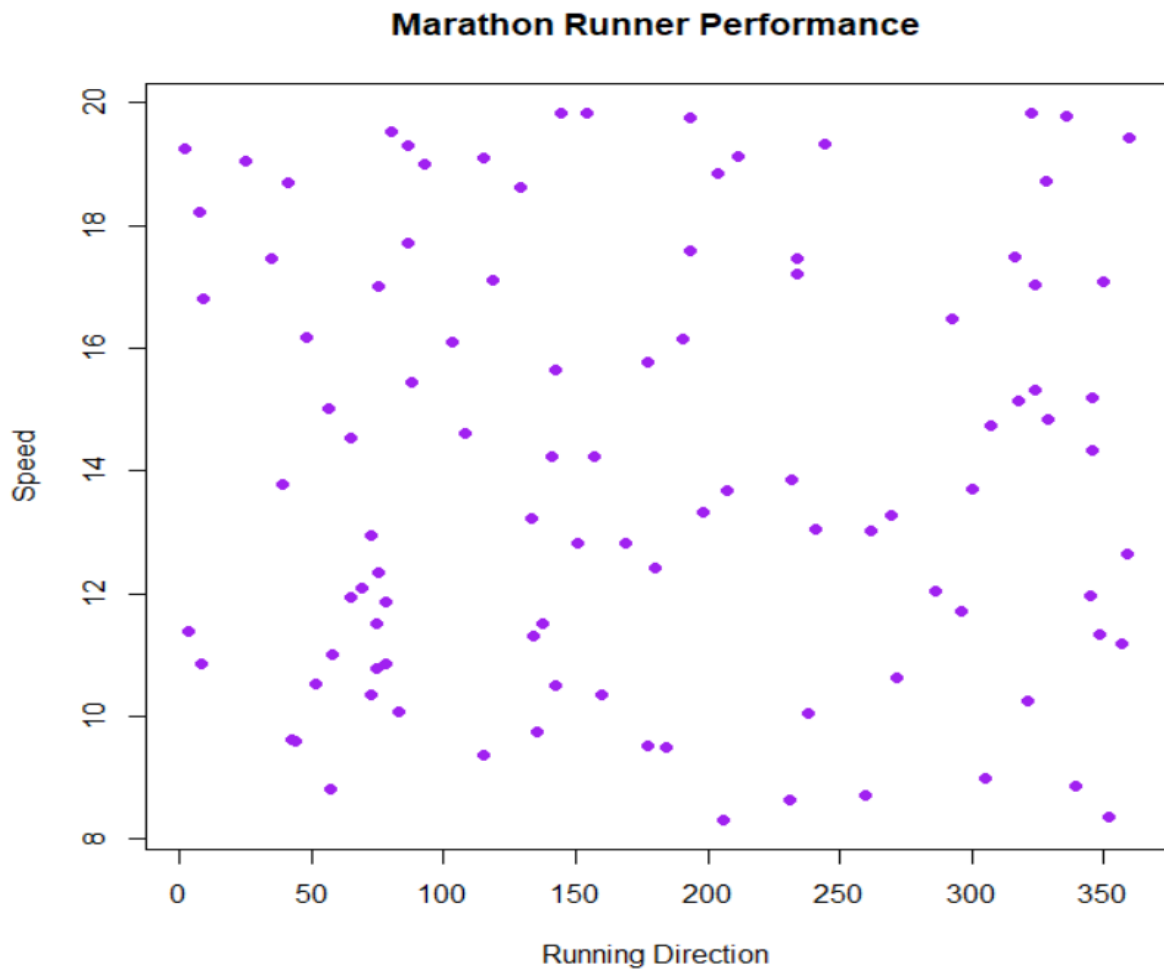
## CODE: R Program

```
set.seed(9)

direction <- runif(100,0,360)
speed <- runif(100,8,20)

plot(direction,
      speed,
      pch=19,
      col="purple",
      xlab="Running Direction",
      ylab="Speed",
      main="Marathon Runner Performance")
```

## OUTPUT:



## Scenario 10: COVID-19 Vaccination Campaign Dashboard

### CODE: Python

```
import numpy as np
import matplotlib.pyplot as plt

districts=["A","B","C","D","E","F","G","H"]

vaccinated=np.random.randint(50,100,8)

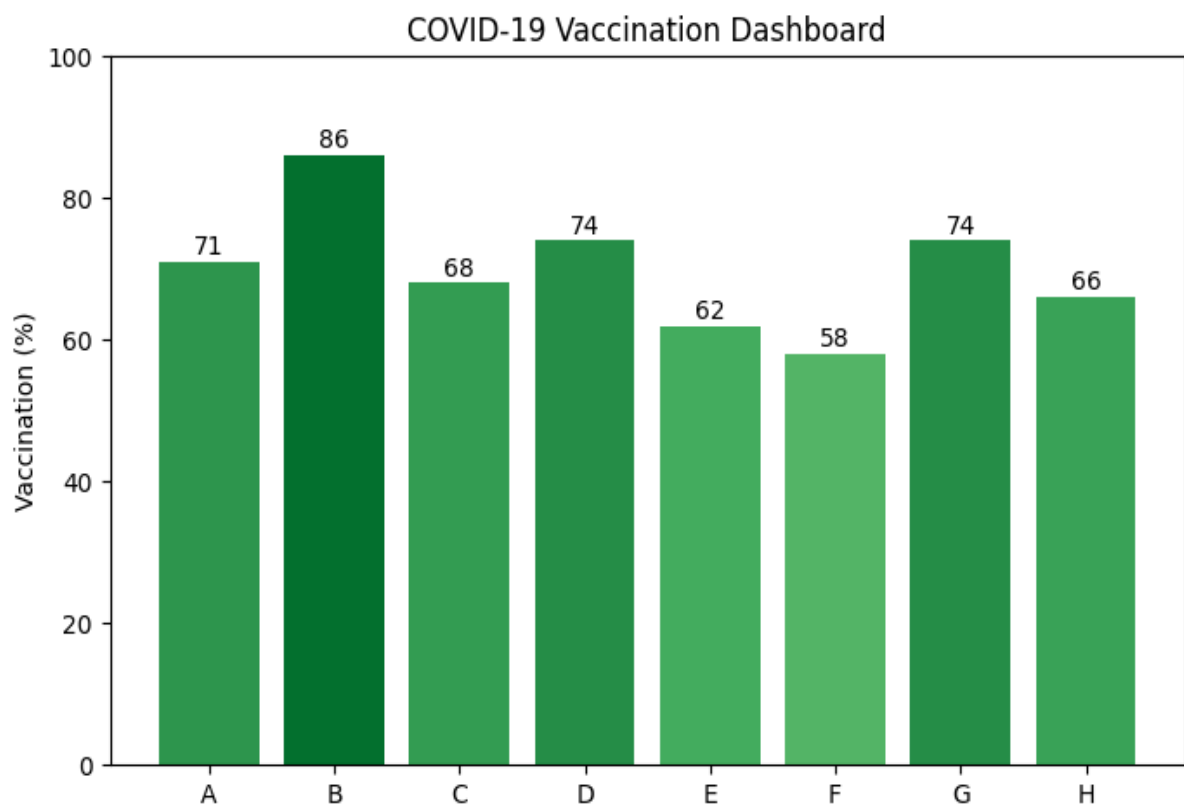
plt.figure(figsize=(8,5))
bars=plt.bar(districts,vaccinated,color=plt.cm.Greens(vaccinated/100))

plt.ylabel("Vaccination (%)")
plt.title("COVID-19 Vaccination Dashboard")
plt.ylim(0,100)

for bar,val in zip(bars,vaccinated):
    plt.text(bar.get_x()+bar.get_width()/2,val+1,str(val),
             ha='center')

plt.show()
```

### OUTPUT:



## CODE: R Program

```
districts <- c("A","B","C","D","E","F","G","H")
vaccinated <- sample(50:100,8)

barplot(vaccinated,
        names.arg=districts,
        col="darkgreen",
        ylim=c(0,100),
        main="COVID-19 Vaccination Dashboard",
        ylab="Vaccination (%)")

text(1:8,
     vaccinated+3,
     labels=vaccinated)
```

## OUTPUT:

